

AL58 - RESEARCH METHODOLOGY AND INTELLECTUAL PROPERTY RIGHTS

UNIT - III DATA COLLECTION, SAMPLING & ANALYSIS WORKSHEET

SECTION 1: DATA COLLECTION METHODOLOGY

Data Type Selection:

- Primary Data: Original data via experiments, direct observation or surveys.
- Secondary Data: Existing data from journals, public records, repositories or reports.

Primary Data Collection Method: (Select and justify one approach)

- Direct Observation: Systematically planned and recorded without asking respondents.
- Personal / Telephone Interview: Structured or unstructured oral-verbal interactions.
- Questionnaire: Self-administered set of questions for large sample sizes.
- Schedules: Structured forms filled in directly by trained enumerators.
- Selection Justification:

Secondary Data Evaluation Criteria: (If using secondary sources, verify these attributes)

- Reliability Assessment: Source credentials and collection time verified.
- Suitability Check: Data aligns with current research scope.
- Adequacy Level: Desired level of accuracy achieved.

SECTION 2: MEASUREMENT SCALES & ERROR CONTROL

Measurement Scale Classification: (Classify your metrics)

- Nominal Scale: Categorical labels (e.g., User Type, Operating System).
- Ordinal Scale: Ordered categories (e.g., Satisfaction Rating, Priority Level).
- Interval Scale: Equal numerical intervals with no true zero (e.g., Temperature, Timestamp Differences).
- Ratio Scale: Numerical scale with an absolute zero (e.g., Response Time in ms, Memory Usage in MB).

Sources of Measurement Error & Mitigation:

- Respondent Bias: Strain or reluctance during responses.
- Situational Factors: Environmental distraction or presence of third parties.
- Measurer Bias: Distortion during wording or recording.
- Instrument Defect: Defective rating scales or measuring tools.
- Mitigation Strategy:

SECTION 3: SAMPLING DESIGN & FRAMEWORK

Target Population vs. Sampling Frame:

- Universe / Population: Entire set of items/units under study.
- Sampling Frame: Specific source list from which the sample is drawn.

Sampling Procedure Selection: (Select representation and selection technique)

- Probability Sampling: Simple Random, Stratified, Cluster or Systematic.
- Non-Probability Sampling: Convenience, Judgment or Quota.

Sampling Parameters & Constraints:

- Sample Size: Total number of units selected.
- Desired Precision Range: Acceptable error bound around estimated parameter.
- Confidence Level: Expected percentage of times parameter falls in precision range.
- Significance Level: Risk of parameter falling outside range.